Classification of Motor Imagery EEG Signals with multi-input Convolutional Neural Network by augmenting STFT

Tanvir Hasan Shovon†, Zabir Al Nazi†, Shovon Dash, Md. Foisal Hossain
Dept. of Electronics and Communication Engineering
Khulna University of Engineering & Technology
Khulna-9203, Bangladesh
tanvirshovon9@gmail.com, zabiralnazi@yahoo.com, shovondash59@gmail.com, foisalkuet@yahoo.com

***Abstract*—Motor imagery EEG classification is a crucial task in the Brain Computer Interface (BCI) system. In this paper, we propose a Motor Imagery EEG signal classification framework based on Convolutional Neural Network (CNN) to enhance the classification accuracy. For the classification of 2 class motor imagery signals, firstly we apply Short Time Fourier Transform (STFT) on EEG time series signals to transform signals into 2D images. Next, we train our proposed multi-input convolutional neural network with feature concatenation to achieve robust classification from the images. Batch normalization is added to regularize the network. Data augmentation is used to increase samples and as a secondary regularizer. A three input CNN was proposed to feed the three channel EEG signals. In our work, the dataset of EEG signal collected from BCI Competition IV dataset 2b and dataset III of BCI Competition II were used. Experimental results show that average classification accuracy achieved was 89.19% on dataset 2b, whereas our model achieved the best performance of 97.7% accuracy for subject 7 on dataset III. We also extended our approach and explored a transfer learning based scheme with pre-trained ResNet-50 model which showed promising result. Overall, our approach showed competitive performance when compared with other methods.**

***Index Terms*—EEG, BCI, STFT, CNN, Augmentation, Transfer Learning.**

I. Introduction

Brain computer interaction is a rapid-growing technology that assists disabled people like paralytic patients and gives solutions to these neurologically disabled patients [1]. Day by day the enthusiasm in this field has dramatically increased. Brain computer interface (BCI) system gives the users with a complete communication system between their body and external devices. It's a two-way process. The motor imaginary process is a mental activity without any actual body movement.

In a study [2], 4 class EEG data was classified and compared with namely minimum distance analysis (MDA), linear discriminant analysis (LDA), k-nearest-neighbor (kNN) classifier and support vector machine (SVM) classifier. Kalman filtering technique was used for parameter extraction. By using an adaptive autoregressive (AAR) theory, the EEG signal was modeled. However, in the part of classification, support vector machine (SVM) performed as the best classifier.

In another work [3], the author proposed a Deep Convolutional Neural Network (DCNN) methodology for identifying 4 movements (left-hand , right-hand , feet and tongue movement), where EEG signals were represented by power spectral density (PSD). Authors proposed method got mean accuracy of 0.8797 ± 0.0296 for the BCI Competition IV Dataset 2a.

In recent year, researchers began to apply Deep Learning for multi-class motor imagery classification. For instance, in [4], the authors proposed a novel method that was robust to unwanted noise and subject variation. In this method, EEG data was converted into multi-dimensional tensor images using Azimuth Equipment Projection (AEP) technique that preserved the spatial, spectral and temporal structure of human brain signals. In the classification part, ConvNets and LSTM network were used that reduced the error rate from 15.3% to 8.9% and increased the accuracy also. One of the main requirement was the information about the coordinates of electrodes in this work. In [5], a new signal to angle-amplitude graph image conversion was proposed. The scale-invariant feature transform (SIFT) technique and the bag of visual words (BoW) were used in the feature extraction stage. In this work, K-NN classifier was employed both for EEG and MEG dataset and performed well. In [6], the authors proposed a method called Weighted Difference of Power Spectral Density (WDPSD) for feature extraction of BCI system based on 2-class motor imagery. Here, the PSD difference matrix of EEG signal was obtained by STFT, CWT and HHT methods form optimal channels. The method experimented on BCI Competition IV Dataset 2a and Dataset 2b and showed good classification accuracy.

In this paper, we propose a multi-input convolutional neural network. Firstly we converted the EEG signals into STFT images. Furthermore, Augmentation was applied to increase the number of images. The performance of this model has been evaluated with two datasets. Such results indicate that proposed multi-input convolutional neural network reduces the classification error and improves the motor imagery EEG

† both authors contributed equally.

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signal classification accuracy.

II. Methods

Methodology section contains STFT, data augmentation and model architecture discussion. Our proposed method is inspired by computer vision and medical image analysis domain. Computer vision filed is flourishing since the development of Convolutional Neural Network. Data Augmentation, Convolutional Neural Network, Feature Concatenation, Transfer learning have been successfully employed in medical signal, and image analysis domain [7–10].

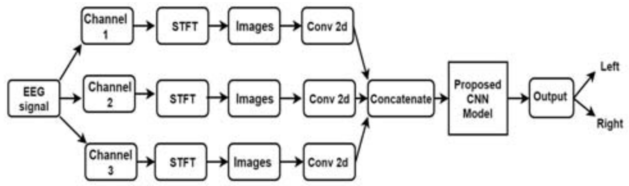

Fig. 1. Basic framework of proposed classification method.

In our approach, EEG signals of Motor Imagery are converted into 2D images with the help of Short Time Fourier Transform. Then the input images are feed into the multi-input Convolutional Neural Network (CNN). The proposed method is evaluated. Fig.1 represents the framework of the proposed method.

A. Short Time Fourier Transform

Short-time Fourier transform (STFT) is a time-frequency analysis used for non-stationary signals analysis. It divides a long time signal into segment with same size of window and apply Fourier transform on it. STFT of signal s(t) is F(τ, ω).

$$F(\tau, \omega) = \int_{-\infty}^{\infty} s(t)h(t-\tau)e^{-j\omega t}dt \tag{1}$$

h(t) is the window function. PSD of $F(\tau, \omega)$ is $P(\tau, \omega)$ which is defined by

$$P_s(\tau, \omega) = \mid F(\tau, \omega) \mid^2 \tag{2}$$

PSD gives a 2D matrix which represents the power of EEG signal with fixed resolution. For better frequency analysis window size is kept wider and on the other hand narrow window is for better time analysis. In our study, STFT was applied on EEG signal. Hanning window was used, where length of each segment is 256.

The EEG signal X is considered to have length d and there are 3 channels. For simplicity, the sampling rate for all channels has been set to f_s. The length of fft is denoted by n_{fft}. The signals are separated channel-wise, and an STFT has been performed for each one.

After STFT, we get the vector of sample frequencies f, vector of segment times t, and the STFT which is of dimension $l_f \times l_t$. We have only considered the absolute value of the STFT, which preserves good amount of information. After taking STFT, we transform the magnitude to RGB color information through function C. Finally, we get RGB images, which are resized to match the input of the CNN. We have used, $m = 224$ so that the transfer learning experiments become simpler.

Algorithm 1 Algorithm for EEG Signal to STFT Image generation

Input: $X_{d\times3}$, f_s, n_{fft}
Output: $I_{m\times m\times3\times3}$

```
X_1 = X_{1:d,1}
X_2 = X_{1:d,2}
X_3 = X_{1:d,3}
I = 0_{m×m×3×3}
for i = 1 to 3 do
    f, t, Z = STFT(X_i, f_s, n_fft)
    Z_r = abs(Z)
    I' = C(Z_r)
    I_i = Resize(I', m × m × 3)
end for
return I
```

B. Data Augmentation

Scarcity of samples problem can be solved by using data augmentation technique that takes images and converts to all the possible form we specify like rotation, flipping, zoom in, zoom out etc. The parameter we have used are rotation of 0 5, flipping probability 0.2, zooming coverage area .08, and brightness min factor 0.7 with max factor 1.3 [11–13]. It is expected that data augmentation will regularize our CNN model and will solve the overfitting problem. Fig.2 represents the augmented images.

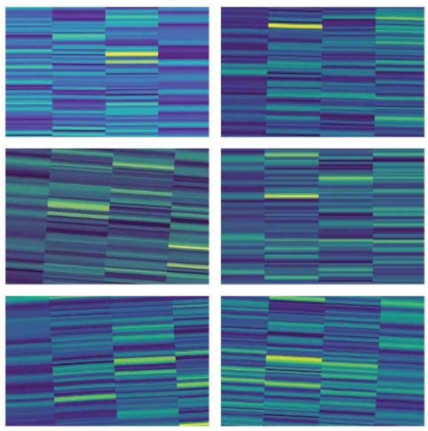

Fig. 2. Augmented images.

C. Model Architecture

Convolutional neural network (CNN) is a special form of deep neural networks. CNN consists of one input layer, one output layer and in between them are multiple convolutional, and dense layers. ConvNet is able to adopt the spatial and temporal information in an image. The objectives of convolution operation is to extract the features. CNN reduces the preprocessing part compared to traditional classification algorithms and hidden layers of CNN carry out the input data in a filtered form. Deeper convolutional layers learn more complex features [14, 15].

We propose a novel methodology for EEG classification with multi-channel CNN and data augmentation to analyze the EEG images. This network has scalable high-performance architecture. Input image size is $(224 \times 224 \times 3)$ for each channel. The input images are convolved with filters in the convolution layers. In this network all convolutional layer use 3×3 size kernel with relu activation and max pooling layer. Max pooling layer minimizes the high dimensional input into smaller ones. The size of window of each max pooling layer is 2×2. Three channel EEG images are used as input of three convolutional layers. Then these convolutional layers were concatenated. Next layer contains 16 kernels of size 3×3 with relu activation layer. After this layer, number of 2D convolutional layers have been applied with number of kernels 16, and 8 respectively. Max pooling layer has been applied after each convolution layer. Batch normalization is added between CNN layers to regularize the CNN model. In later experiments, batch normalization layers were removed as it helped to decrease the loss quickly. Finally, a fully connected layer is attached on the top of the convolutional layer with 8 neurons. The last layer is softmax classification layer. The model has in total only 573,704 parameters, so it's easy and very fast to train. The proposed model architecture is illustrated in Fig.3.

TABLE I
ARCHITECTURE OF THE CNN MODEL

Layer	Output tensor shape	# parameters
Input 1	$B\times224 \times 224 \times 3$	0
Input 2	$B\times224 \times 224 \times 3$	0
Input 3	$B\times224 \times 224 \times 3$	0
Conv2D 1	$B\times224 \times 224 \times 32$	896
Conv2D 2	$B\times224 \times 224 \times 32$	896
Conv2D 3	$B\times224 \times 224 \times 32$	896
Concatenate 1	$B\times672 \times 224 \times 32$	0
Conv2D 4	$B\times670 \times 222 \times 16$	4624
Activation 1	$B\times670 \times 222 \times 16$	0
Conv2D 5	$B\times668 \times 220 \times 16$	2320
Activation 2	$B\times668 \times 220 \times 16$	0
Max_pooling 1	$B\times334 \times 110 \times 16$	0
Conv2D 6	$B\times332 \times 108 \times 8$	1160
Max_pooling 2	$B\times166 \times 54 \times 8$	0
Flatten	$B\times71712$	0
Dense 1	$B\times8$	573704
Dense 2	$B\times2$	18

* Here, B denotes the batch size.

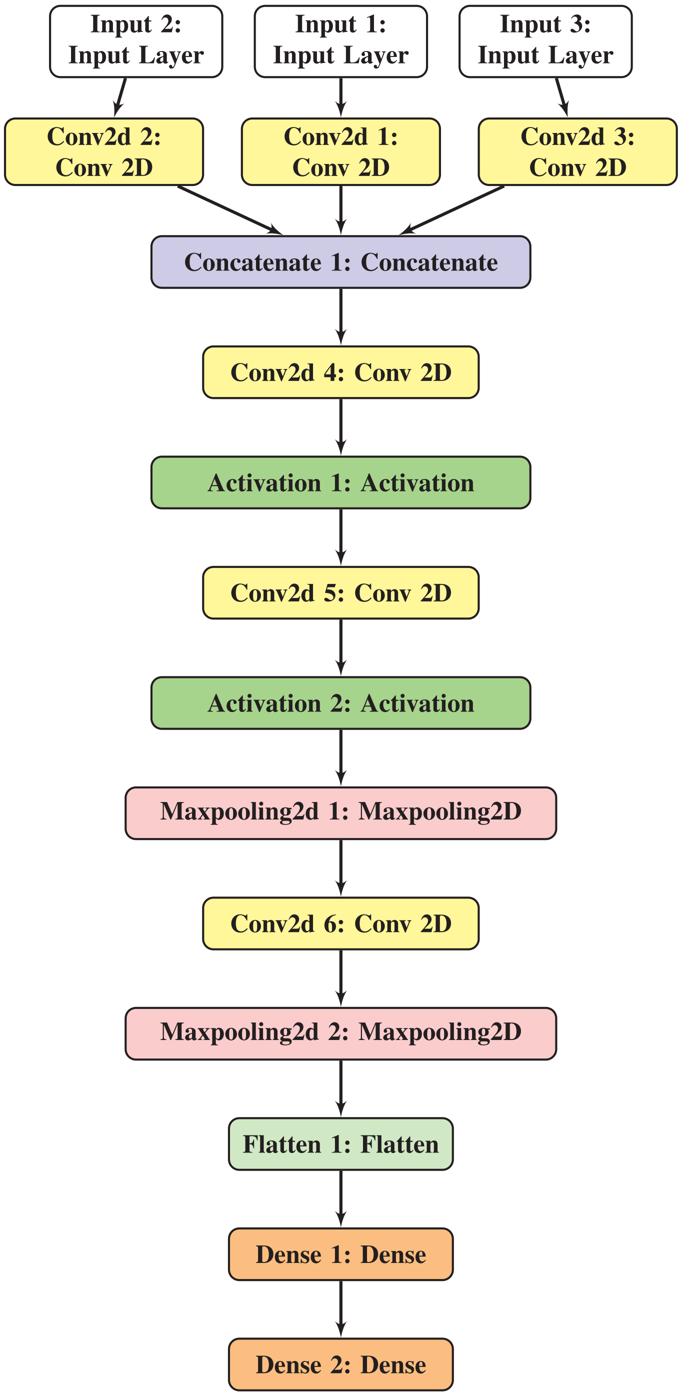

Fig. 3. CNN Architecture

III. Experiments And Results Analysis

A. Dataset

The EEG dataset used in this work were dataset III from BCI Competition 2 [16] and training part of dataset 2b from BCI Competition IV [17]. Both datasets contain EEG data for Motor Imagery task. EEG was collected form 3 channels C3, Cz and C4. Table II describes the information of datasets.

Dataset 2b of BCI Competition 4 includes three sessions as training set. Third session includes online feedback at a 250 Hz sampling frequency and other two sessions are without feedback. Per session resulted in 20 trials per run and 120 trials in total. Visual cue is displayed for 1.25s. After that the subjects perform 4s motor imagery task. In the Dataset III of BCI competition 2, Dataset sampling frequency was 128 Hz. The data set forms of 140 trials for training set. The length of recording per trial was 9s.

TABLE II
DATASET DESCRIPTIONS

Dataset	Subjects	Channels	Trials	Rate
Competition 2 dataset III [16]	1	C3, Cz and C4	280	128
Competition 4 dataset IIb [17]	9	C3, Cz and C4	400	250

B. Experimental Study

Transfer learning is a learning technique in machine learning domain, where a model learned on one task is re-purposed on a second task. It is an optimization technique which improves the performance of the model. In transfer learning, firstly we train a network on specific dataset then, after learning the features we transfer them to a second network to train on another target dataset.

TABLE III
ACCURACY RESULT OF TRANSFER LEARNING PROCESS

Subject	Classifier	Accuracy(%)	Recall(%)	Precision(%)
S1	SVM	93.33	96.33	88.98
	Random Forest	80.35	77.98	73.84
	Decision tree	79.91	82.56	72.54
	KNN	86.61	89.91	80.21
S2	SVM	94.64	95.412	91.63
	Random Forest	83.48	83.48	77.10
	Decision tree	76.78	79.81	69.17
	KNN	86.60	91.74	79.83

ResNet is a class of Residual Networks. In the residual learning, we try to learn some residual,in lieu of learning some features. ResNet introduced the concept of skip connections. Residual can be easily realized as elimination of feature which has learned from any layer and the input of that layer. ResNet makes this using shortcut connections (straightly connecting input of nth layer to some $(n + x)^{th}$ layer. The ResNet-50 model made up of 5 stages, each has a convolution and Identity block. Each convolution block has 3 convolution layers and each identity block also has 3 convolution layers. The ResNet-50 has over 23 million trainable parameters.

We used pre-trained ResNet-50 weight, trained on ImageNet natural image dataset, and used it as a feature extractor for our EEG STFT images [18]. After feature extraction, we compared multiple classifiers for EEG classification. It can be observed from Table III, the results are very promising. So, the features were useful for EEG STFT images, even though the ResNet-50 model was trained on natural images. This experiment demonstrates the greater applicability and scopes for transfer learning.

The t-SNE embedding for the extracted features by ResNet-50 model is shown in Fig.4. Small clusters formed by two classes can be observed in the embedding plot.

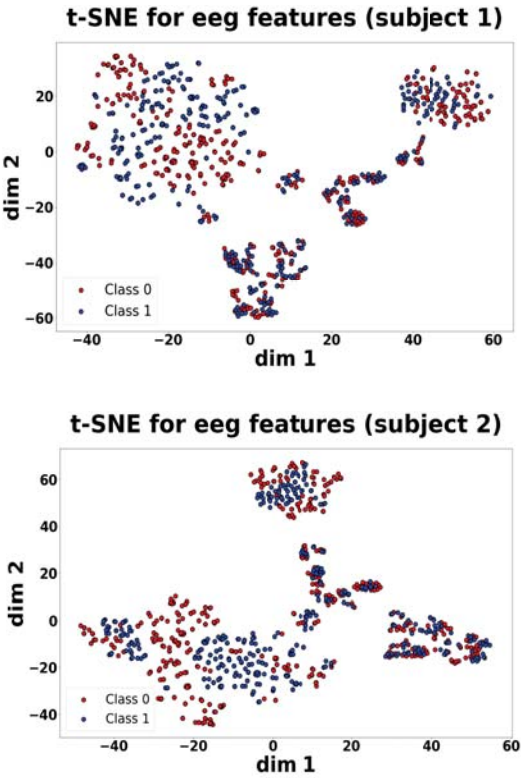

Fig. 4. t-SNE embedding for subject 1 and 2.

C. Results

In this study, the proposed method was experimented on BCI Competition IV dataset 2b. Additional 1000 augmented images were generated. The images were split into train and test images for each subject. For each subject performance was evaluated with four performance metrics: Accuracy, Precision, Sensitivity, and Specificity. Table IV shows that for subject 1, 4 and 7 where our model achieved highest accuracy compared to [19] but for subject 2 our model performed poorly in BCI Competition 4 dataset 2b. Average accuracy for all subjects was 89.19%. In each session 90% images were selected for training and 10% were selected for testing. We used 32 epochs with batch size 32 for training our model. Table V represents the result of dataset III of BCI Competition II We compared our results with other works based on same dataset. The authors in [19] proposed CNN and Stacked Autoencoders (SAE) based deep learning method to classify 2 class motor imagery signals. In study [20], sparse kernel machine was proposed to classify left and right hand signals.

In order to evaluate our model we compare our results with 2 other recent works. Table VII represents the comparison for dataset 2b of BCI Competition IV. For all 9 subjects the performance of our model is better than other results. Though subject 2 performance accuracy is not high but still competitive. In reference [19], author proposed sparse kernel mechanism and got highest accuracy for subject 4 and 9. On the other hand In reference paper [20] author proposed

TABLE IV
ACCURACY RESULTS OF SESSION-TO-SESSION CLASSIFICATION FOR BCI COMPETITION IV DATASET 2B.

Subject	Accuracy(%)	Precision(%)	Sensitivity(%)	Specificity(%)
1	93.36	93.91	93.10	93.64
2	63.33	68.38	61.29	65.61
3	94.7	93.33	96.55	92.86
4	96.68	98.21	95.65	98.17
5	84.76	85.46	84.29	83.62
6	93.75	97.39	91.06	97.03
7	97.77	97.35	98.21	97.32
8	91.09	87.90	94.4	87.93
9	95.09	94.74	95.58	94.59

a deep learning approach named CNN-SAE model and got highest accuracy for subject 4. For all other cases, our model outperformed those approaches.

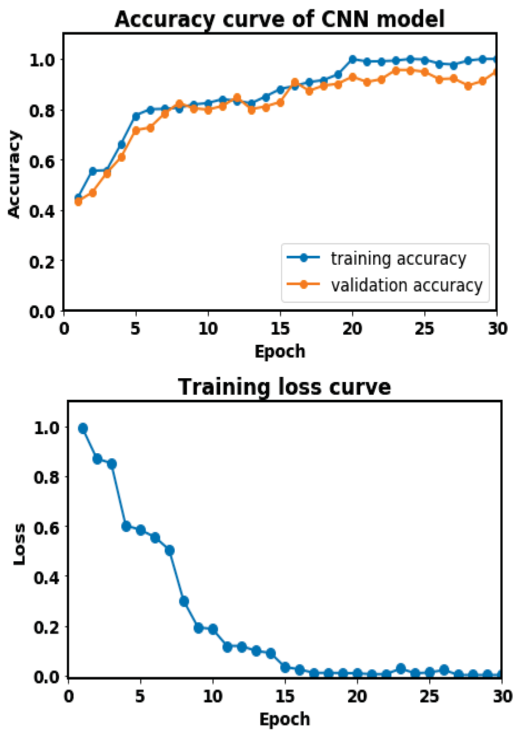

Fig. 5. Accuracy and loss curve for subject 4.

From Fig.5, the convergence of model after epoch 16 can be observed, the training and validation accuracy plots are also very close, so the model shows no overfitting on training data. And training loss curve decreases after increasing number of epochs.

TABLE V
RESULT TABLE FOR DATASET III OF BCI COMPETITION II.

Dataset	Accuracy(%)	Precision(%)	Sensitivity (%)	Specificity (%)
BCI II	89.73	90.09	89.29	90.18

The comparison results for BCI Competition II dataset III are shown in table VI. In our case, the proposed model accuracy is 89.73% which is better than reference papers.

TABLE VI
ACCURACY COMPARISON TABLE WITH OUR PROPOSED MODEL TO RECENT WORKS FOR DATASET III OF BCI COMPETITION II.

Dataset	Accuracy (%) [Proposed method]	Accuracy (%) [19]	Accuracy (%) [12]
BCI II	89.73	90.0	88.2

The result shows that, the proposed method is more reliable and robust for motor imagery EEG classification.

TABLE VII
ACCURACY COMPARISON TABLE WITH OUR PROPOSED MODEL TO RECENT WORKS FOR DATASET 2B OF BCI COMPETITION IV.

Subject	Accuracy (%) [Proposed method]	Accuracy (%) [19]	Accuracy (%) [11]
1	93.08	78.1	75.94
2	63.33	63.1	61.79
3	94.74	60.6	61.25
4	96.88	95.6	95.63
5	84.76	78.1	93.75
6	93.75	73.8	84.38
7	97.76	70	77.81
8	91.09	70.6	91.25
9	95.09	82.5	87.19

IV. CONCLUSION

In this work, we proposed multi-input CNN to classify multi-channel EEG Motor Imagery Signals which is an important and novel approach towards Brain Computer Interface Research. Our method uses computer vision techniques in time series analysis and showed good performance. Here, we use both the time-domain and frequency domain information from the EEG as input. As, STFT uses coarse representation of the EEG, we can control the resolution to scale the network. As, increasing sampling rate increases the length of input EEG signals, taking the STFT will be helpful. STFT was utilized to generate images from time-series multi EEG data. Augmentation was useful to increase number of training samples which is mandatory to train Convolutional Neural Network in our case due to data scarcity. Finally, the model was evaluated on data from multiple subjects and showed good performance. This work will be useful in Brain Computer Interface and time series analysis research.

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